GLOBAL FUEL PRICES ANALYSIS

Project Documentation

Prepared by: Vidhya Jain
Tool: Tableau Public | Supporting files: GitHub repository

1. Project Overview

Global Fuel Prices Analysis is a data-visualization project that explores differences in petrol prices across countries and examines how fuel prices relate to factors such as taxes and crude oil prices. The project uses Tableau dashboards and a Story to present the analysis in a visual and accessible way.

2. Problem Statement

Fuel prices can vary considerably between countries. Looking at raw data alone can make it difficult to compare countries, identify broad patterns, and understand how taxation and crude oil prices may relate to retail petrol prices. This project organizes the available data into visualizations to support easier comparison.

3. Objectives

- Explore and compare petrol prices across countries using the available dataset.
- Present price differences through charts and interactive dashboards.
- Examine tax levels and their relationship to fuel prices where the dataset provides the required fields.
- Explore the relationship between petrol prices and Brent crude oil prices.
- Publish the Tableau Story online and provide supporting project files through GitHub.

4. Dataset and Tools

The project uses the cleaned fuel-price dataset prepared for the analysis. The files used during the project include a CSV dataset and an Excel dataset. Confirm the exact original dataset source and add its URL below before final submission.

Dataset file(s): fuel_prices_clean.csv; global_fuel_prices_Dataset.xlsx

Original dataset source / URL: [Add the source link used for the dataset]

Tools used:

- Tableau Desktop / Tableau Public for data visualization, dashboards, and Story publishing.
- CSV and Excel files for storing and preparing the data.
- GitHub for hosting project files and the README.

5. Data Preparation

The dataset was prepared for visualization before building the charts. The cleaned CSV and Excel project dataset were used as the data inputs. Review the final workbook and add any specific cleaning steps that were actually performed (for example, handling missing values, correcting field types, or removing duplicates).

6. Calculated Fields

The following calculated fields were created in Tableau:

- Tax Band: groups tax percentage into High (35% or above), Medium (20% to below 35%), and Low (below 20%).
- Price Category: groups Petrol Usd Liter into Expensive (3 or above), Medium (1 to below 3), and Cheap (below 1).
- Price Brent Ratio: compares average petrol price with average Brent crude price using the formula AVG([Petrol Usd Liter]) / AVG([Brent Crude Usd]) * 100.

Note: Verify the field names and thresholds against the final Tableau workbook before submission.

7. Visualizations and Dashboards

Eight charts were created for the project. The visualizations are arranged into dashboards intended to present different perspectives on global fuel prices. The project includes dashboards identified during development as:

- Dashboard 1 – Policy: supports comparison and exploration of fuel-price and tax-related patterns.
- Dashboard 2 – Logistics: presents fuel-price information from a logistics or operational perspective.
- Dashboard 3 – Research: supports deeper exploration of price patterns and related measures.

Add or update the exact chart titles and screenshots from the final Tableau workbook so this section matches the published version.

8. Filters and Interactivity

Where configured in the final workbook, dashboard filters allow users to narrow the displayed data and explore specific countries or categories. Confirm which filters are active in the published dashboards and list them here.

9. Key Insights

The dashboards are designed to help users compare petrol prices across countries, review differences between tax bands and price categories, and explore the relationship between retail petrol prices and Brent crude prices. Insert two or three specific findings observed from the final charts before submission; do not report numerical conclusions unless they are verified from the workbook.

10. Publishing and Project Links

Tableau Public Story:

https://public.tableau.com/app/profile/vidhya.jain/viz/GlobalFuelPricesAnalysis/Story1?publish=yes

GitHub repository: [https://github.com/vidhya-jain/fuel-prices-project]

Demo / explanation video:
[https://drive.google.com/file/d/1OFsIwGC01hxrHGgM_6wzWetBCvSMjlsk/view?usp=sharing]

11. Conclusion

This project demonstrates how Tableau can be used to transform a fuel-price dataset into interactive visual analysis. The charts, dashboards, calculated fields, and published Story provide a structured way to explore differences in petrol prices and related factors. The project can be improved further by adding verified findings, documenting the exact dataset source, and including the final demonstration video.

12. Final Submission Checklist

☐ ✓ Confirm the original dataset source URL.

☐ ✓ Verify calculated-field formulas and thresholds in Tableau.

☐ ✓ Add exact chart names and screenshots from the final dashboards.

☐ ✓ Write two or three verified insights from the visualizations.

☐ ✓ Insert the final GitHub repository URL.

☐ ✓ Record and add the project explanation video link if required.

☐ ✓ Check that the published Tableau Story opens correctly.